Coordinates & Transformations

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: A point A is at (-3, 5). What is its x-coordinate and what is its y-coordinate?
Method & answer: x-coordinate: -3. y-coordinate: 5. It is in the top-left region of the grid because x is negative and y is positive.

Standard: A point B is at (-2, -4). It is translated 5 right and 3 up. What are its new coordinates?
Method & answer: 5 right: -2 + 5 = 3. 3 up: -4 + 3 = -1. New point: (3, -1).

Advanced: A point C is at (3, -5). It is rotated 90° clockwise about the origin, using the rule (x, y) → (y, -x). What are its new coordinates?
Method & answer: (y, -x) = (-5, -3).

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point A is at (-4, 6). What is its x-coordinate?

2. A point A is at (-4, 6). What is its y-coordinate?

3. A point starts at (-2, 3) and is translated 5 right. What are its new coordinates?

4. A point starts at (4, -6) and is translated 9 up. What are its new coordinates?

5. A point is at (-5, 2). It is translated 3 right. What are its new coordinates? A) (-2, 2) B) (-8, 2) C) (-5, 5) D) (-5, -1)

6. A point is at (-3, -3). Which quadrant of the grid is it in?

7. A point is at (5, -2). Which quadrant of the grid is it in?

8. A point starts at (-6, 2) and is translated 4 right and 5 down. What are its new coordinates?

9. A point starts at (3, 8) and is translated 7 left and 2 down. What are its new coordinates?

10. Select ALL that apply. Look at the points P(-2, 4), Q(2, -4), R(-2, -4) and S(2, 4). Which do NOT lie in the third quadrant? A) P(-2, 4) B) Q(2, -4) C) R(-2, -4) D) S(2, 4)

11. A point starts at (0, -5) and is translated 3 left and 3 up. What are its new coordinates?

12. A point starts at (-4, -4) and is translated 6 right and 6 up. What are its new coordinates? A) (2, 2) B) (-10, -10) C) (2, -10) D) (-10, 2)

13. A point is at (-6, 3). It is reflected in the x-axis. What are its new coordinates?

14. A point starts at (7, -1) and is translated 9 left and 4 up. What are its new coordinates?

15. Which of these points lies exactly on the y-axis? A) (0, -7) B) (-7, 0) C) (7, 7) D) (-7, -7)

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point is at (-4, 5). It is reflected in the x-axis. What are its new coordinates?

6. A point is at (-8, -4). It is reflected in the y-axis. What are its new coordinates?

11. Select ALL that apply. A rectangle has vertices at (-3, -2), (5, -2), (5, 4) and (-3, 4). Which statements are true? A) Its width is 8 units B) Its height is 6 units C) The vertex (5, 4) is diagonally opposite (-3, -2) D) All four vertices

have a positive y-coordinate

2. A point is at (6, -3). It is reflected in the y-axis. What are its new coordinates?

3. A point starts at (-3, -3) and is translated 8 right and 8 up. What are its new coordinates?

4. A point at (-6, -2) is reflected in the x-axis. What are its new coordinates? A) (-6, 2) B) (6, -2) C) (6, 2) D) (-6, -2)

5. A point is at (5, 4). It is reflected in the x-axis, then the result is reflected in the y-axis. What are the final coordinates?

7. A line segment joins A(-4, 3) and B(4, 3). What is the length of AB?

8. A point starts at (9, -7) and is translated 12 left and 5 up. What are its new coordinates?

9. A point at (7, -5) is reflected in the y-axis. What are its new coordinates? A) (-7, -5) B) (7, 5) C) (-7, 5) D) (5, -7)

10. A point is at (-9, -6). It is reflected in the x-axis. What are its new coordinates?

12. A point starts at (-1, -1) and is translated 10 right and 10 up. What are its new coordinates?

13. A point at (-2, 6) is reflected in the x-axis to give P, then P is translated 3 down. What are the final coordinates?

14. A point starts at (-5, 6) and is translated 8 right and 9 down. What are its new coordinates? A) (3, -3) B) (-13, -3) C) (3, 15) D) (-13, 15)

15. A point starts at (0, 0) and is translated 7 left and 7 down. What are its new coordinates?

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point is at (2, 5). It is rotated 90° clockwise about the origin. What are its new coordinates?

2. A point is at (4, -3). It is rotated 90° anticlockwise about the origin. What are its new coordinates?

3. A point is at (-3, 6). It is rotated 180° about the origin. What are its new coordinates?

4. A point at (5, 2) is rotated 180° about the origin. What are its new coordinates? A) (-5, -2) B) (5, -2) C) (-5, 2) D) (2, 5)

5. A point is at (0, -6). It is rotated 90° clockwise about the origin, using (x, y) → (y, -x). What are its new coordinates?

6. A point starts at (-4, -8) and is translated 12 right and 3 up. What are its new coordinates?

7. A square has vertices at (-2, -2), (2, -2), (2, 2) and (-2, 2). What is its side length, and what are the coordinates of its centre?

8. A point is at (-3, -5). It is rotated 90° anticlockwise about the origin. What are its new coordinates?

9. A point is at (-7, 9). It is reflected in the x-axis. What are its new coordinates?

10. Select ALL that apply. A shape has vertices A(-1, 0), B(-1, 4), C(3, 4) and D(3, 0). It is rotated 180° about the origin. Which of these are correct new vertices? A) A becomes (0, -1) B) B becomes (1, -4) C) C becomes (-3, -4) D) D becomes (-3, 0)

11. A point starts at (6, -9) and is translated 14 left and 15 up. What are its new coordinates?

12. A point at (4, -6) is reflected in the y-axis, and the result is then rotated 90° clockwise about the origin using (x, y) → (y, -x). What are the final coordinates?

13. A rectangle has vertices at (-5, -3), (1, -3), (1, 2) and (-5, 2). What is its perimeter? A) 22 units B) 11 units C) 18 units D) 30 units

14. A point is at (-1, -9). It is reflected in the y-axis. What are its new coordinates?

15. A point starts at (-6, -6) and is translated so that it ends up at (0, 0). What was the translation?